Grade yourself: formula written? numbers substituted? arithmetic shown? answer boxed to 4 decimals? interpretation sentence where asked? Each part on the real exam is scored on those steps, not only the final number.

Question 1 (6 points) — Commuting and Housing

First add the margins: Drive 42, Bus 26, Walk 26, Bike 16 (row totals); on campus 49, off campus 61; overall 110.

(a) one category → its total ÷ grand total

$$P(\text{off campus}) = \frac{61}{110} = 0.5545$$

0.5545

(b) a row → add its cells, ÷ 110

$$P(\text{walk}) = \frac{22 + 4}{110} = \frac{26}{110} = 0.2364$$

0.2364

(c) OR → add, subtract the overlap (drive AND on campus = 8) once

$$P(\text{drive or on}) = \frac{42}{110} + \frac{49}{110} - \frac{8}{110} = \frac{83}{110} = 0.7545$$

0.7545

(d) AND → the one cell ÷ 110

$$P(\text{bike and off}) = \frac{7}{110} = 0.0636$$

0.0636

(e) GIVEN off campus → divide by the off-campus total (61), not 110

$$P(\text{bus} \mid \text{off}) = \frac{16}{61} = 0.2623$$

0.2623

Common errors: (c) forgetting the overlap gives 91/110 = 0.8273; (e) 16/110 = 0.1455 uses the wrong denominator.

Question 2 (2 points) — Valid distribution?

Rule 1: 0.15, 0.25, 0.30, 0.20, 0.12 are all between 0 and 1 ✓. Rule 2: add them.

$$0.15 + 0.25 + 0.30 + 0.20 + 0.12 = 1.02 \neq 1$$

NOT a valid probability distribution — the probabilities add to 1.02, not 1.

Write both checks; the question says "explain why."

Question 3 (8 points) — Cars per Household

(a) P(x) = frequency ÷ total; total = 13 + 29 + 33 + 10 = 85

$$P(0)=\frac{13}{85}=0.1529 \qquad P(1)=\frac{29}{85}=0.3412 \qquad P(2)=\frac{33}{85}=0.3882 \qquad P(3)=\frac{10}{85}=0.1176$$

Check: $0.1529+0.3412+0.3882+0.1176=0.9999\approx 1$ ✓ (rounding)

(b) x across (0, 1, 2, 3), P(x) up; a vertical bar above each x with heights 0.1529, 0.3412, 0.3882, 0.1176; tallest at x = 2; label both axes.

(c)

$$\mu=\sum x\,P(x)=0(0.1529)+1(0.3412)+2(0.3882)+3(0.1176)$$
$$=0+0.3412+0.7764+0.3528=1.4704\approx 1.47$$

μ ≈ 1.47 cars

Interpretation: over many households in this neighborhood, the average number of cars per household is about 1.47.

(d) $\sigma=\sqrt{\Sigma(x-\mu)^2\cdot P(x)}$ with μ = 1.47

$$(0-1.47)^2(0.1529)=2.1609(0.1529)=0.3304 \qquad (1-1.47)^2(0.3412)=0.2209(0.3412)=0.0754$$
$$(2-1.47)^2(0.3882)=0.2809(0.3882)=0.1090 \qquad (3-1.47)^2(0.1176)=2.3409(0.1176)=0.2753$$
$$\sigma^2=0.3304+0.0754+0.1090+0.2753=0.7901 \qquad \sigma=\sqrt{0.7901}=0.89$$

σ ≈ 0.89 cars

(Unrounded from the raw counts: μ = 1.4706, σ² = 0.7903, σ = 0.8890; the same to 2 decimals.)

Question 4 (8 points) — Reading Habits (binomial, n = 8, p = 0.35, 1 − p = 0.65)

$$P(x)={}_8C_x\,(0.35)^x(0.65)^{8-x}$$

(a) exactly 2

$${}_8C_2=\frac{8\cdot 7}{2\cdot 1}=28 \qquad P(2)=28(0.35)^2(0.65)^6=28(0.1225)(0.075419)=0.2587$$

0.2587

(b) at most 2 → x = 0, 1, 2 (the 2 is IN)

$$P(0)=(0.65)^8=0.0319 \qquad P(1)=8(0.35)(0.65)^7=0.1373 \qquad P(2)=0.2587$$
$$P(X\le 2)=0.0319+0.1373+0.2587=0.4279\approx 0.4278$$

0.4278 (0.4279 if you add the rounded values — both accepted)

(c) at least 1 → 1 − P(0)

$$P(X\ge 1)=1-(0.65)^8=1-0.0319=0.9681$$

0.9681

(d) n = 300

$$\mu=np=300(0.35)=105 \qquad \sigma=\sqrt{np(1-p)}=\sqrt{300(0.35)(0.65)}=\sqrt{68.25}=8.26$$

μ = 105, σ ≈ 8.26

Exact values: P(2) = 0.2587, P(0) = 0.0319, P(1) = 0.1373, P(X ≤ 2) = 0.4278.

Question 5 (5 points) — Shuttle Wait Time (uniform, a = 0, b = 25)

(a)

$$\text{height} = \frac{1}{b-a} = \frac{1}{25-0} = \frac{1}{25} = 0.04$$

a flat line (rectangle) at height 1/25 from 0 to 25 minutes, and 0 elsewhere; area = 25 × 1/25 = 1

(b)

$$P(5 \le X \le 15) = \frac{15-5}{25-0} = \frac{10}{25} = 0.4000$$

0.4000

(c) "more than 18" → from 18 up to the top end 25

$$P(X > 18) = \frac{25-18}{25-0} = \frac{7}{25} = 0.2800$$

0.2800

Question 6 (6 points) — Normal Distribution and z-Scores

(a) area to the LEFT of a negative z: the hint $P(Z < -0.85) = 0.1977$ is already the answer (no $1 -$ needed)

$$P(Z < -0.85) = 0.1977$$

0.1977

(b) between = (left of the bigger z) − (left of the smaller z); -0.25 is bigger than -1.30

$$0.4013 - 0.0968 = 0.3045$$

0.3045

(c)

$$z = \frac{300-250}{40} = 1.25 \;\rightarrow\; \text{left} = 0.8944 \qquad P(X > 300) = 1 - 0.8944 = 0.1056$$

0.1056

Question 7 (8 points) — Heights of Adult Men (μ = 69, σ = 2.8)

(a) taller than 74 → right tail

$$z = \frac{74-69}{2.8} = 1.79 \;\rightarrow\; \text{left} = 0.9633 \qquad P(X > 74) = 1 - 0.9633 = 0.0367$$

0.0367

(b) shorter than 65 → left area (the hint is already the left area)

$$z = \frac{65-69}{2.8} = -1.43 \qquad P(X < 65) = 0.0764$$

0.0764

(c) between 66 and 72

$$z_1 = \frac{66-69}{2.8} = -1.07 \rightarrow 0.1423 \qquad z_2 = \frac{72-69}{2.8} = 1.07 \rightarrow 0.8577$$

$$P(66 < X < 72) = 0.8577 - 0.1423 = 0.7154$$

0.7154

(d) 10th percentile $\rightarrow$ area to the LEFT $= 0.10 \rightarrow$ hint z $= -1.28$ (negative: lower tail)

$$x = \mu + z\sigma = 69 + (-1.28)(2.8) = 69 - 3.584 = 65.416 \approx 65.4$$

$\approx$ 65.4 inches

Sanity checks: (a) must be below 0.5 (74 is above average); (d) must be below 69.